

# OverSoc

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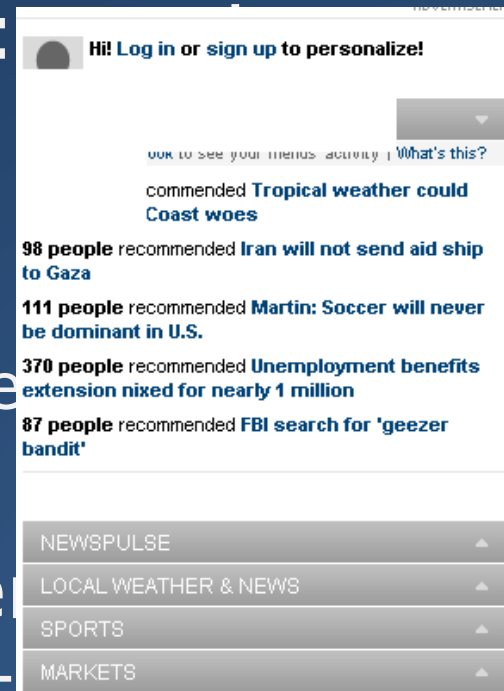
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# Issues

- Social networking profiles are the ultimate means to make targeted advertisements

- Targets situ

life:



- Foll

- Facebook's beacon
    - "Personalize" websites (example)

- Access control is difficult

- OSNs favor open models, users can explicitly disable content or opt out

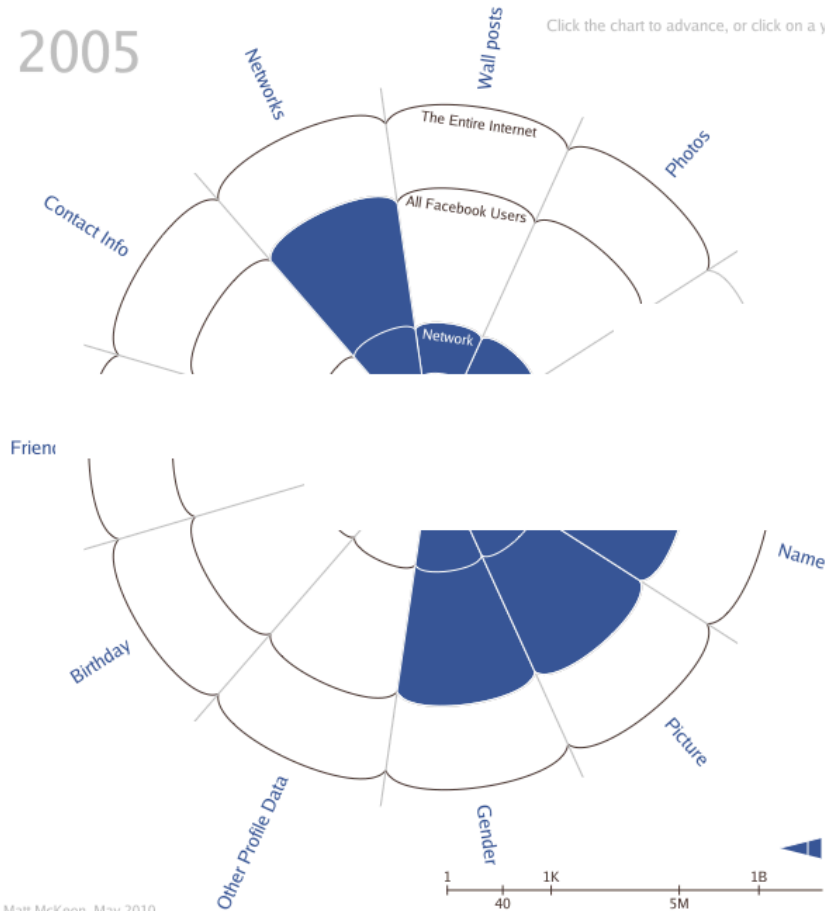
OSNs, overtime, have made more

private information public

# Privacy in Facebook

2005

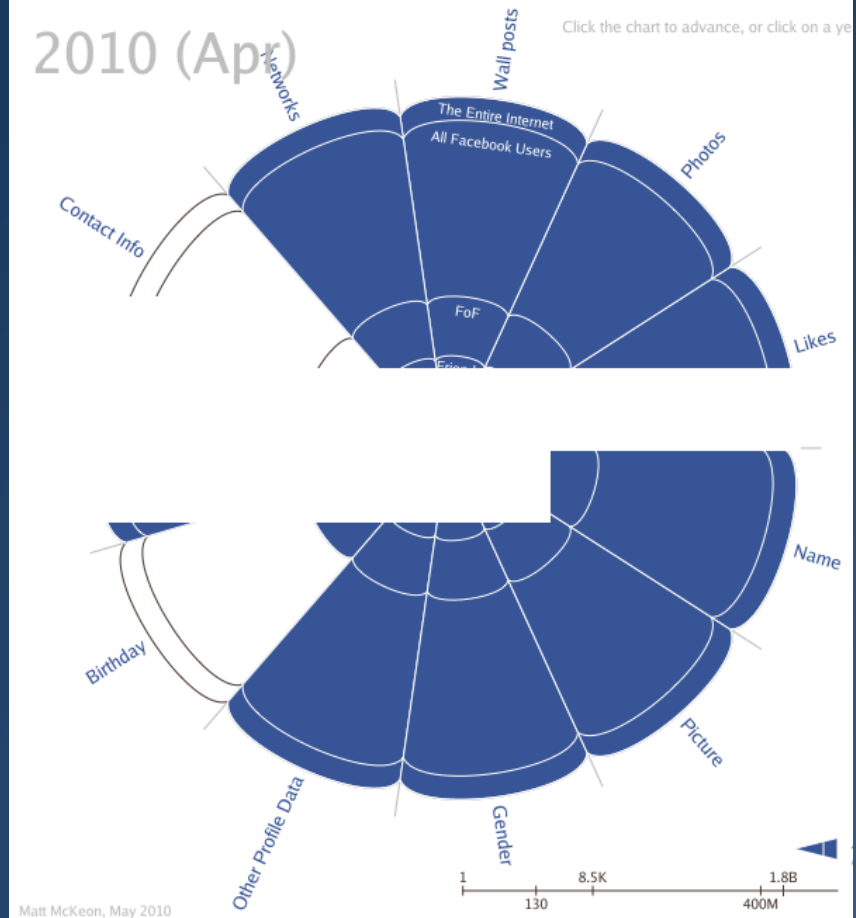
Click the chart to advance, or click on a year



Matt McKeon, May 2010

2010 (Apr)

Click the chart to advance, or click on a year



Matt McKeon, May 2010

# Motivation

- Centralized services require capital to continue, motivating OSNs to be invasive
- Users can change their behavior, using different, more decentralized services
- Many ways to organize a decentralized OSN, existing work:
  - Requires a user to be online to be accessible
  - Replicate entire profiles to many users
  - Challenges in finding friends
- Structured overlays can be leveraged

# Outline

- Issues and Motivation
- OverSoc Introduction
- State of Str                      rlays
- 
- Rem

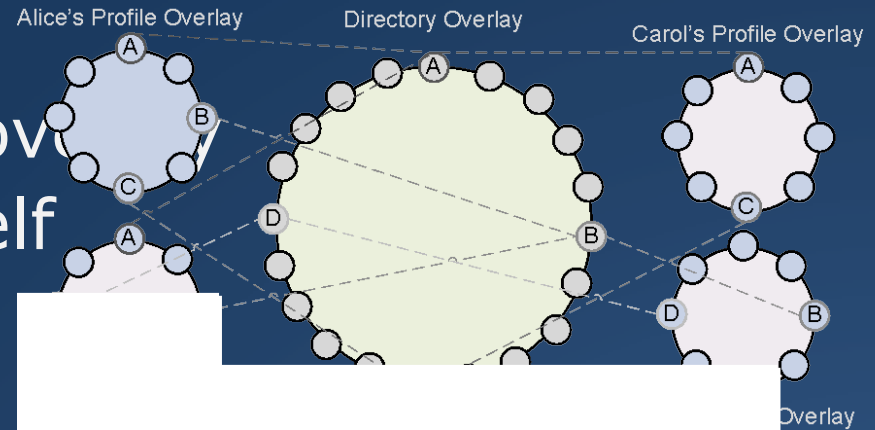
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# Introducing OverSoc

- User-centric

- Each has their own overlay
- Content stored on self and peers



- All I  
authorized peers can access

d, only

- Share a common directory overlay

- Used to find peers
- Used to connect to users' overlays

- Groups – shared user overlays

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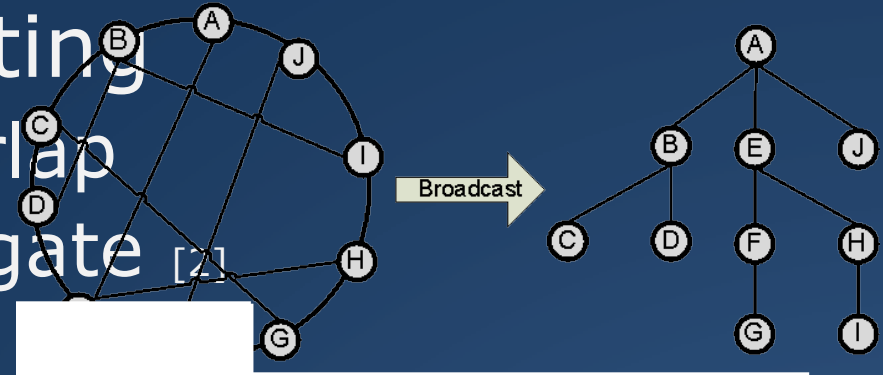


# Structured Overlays

- Efficient key/value lookup through DHT
    - $O(\log N)$  look up time
    - Results from eDonkey KAD studies show that DHT data sharing
  - $t$ 
    - Replicate sufficiently (10 times)
  - Real world usage
    - eDonkey KAD finds media via hashes / keywords
    - Torrentless BitTorrent Tracker
    - LimeWire to assist in firewall / NAT
-  (push proxies)

# Structured Overlays

- Efficient broadcasting
  - $\log^2(n)$ , no overlap
  - Distribute / aggregate
    - First broad



r

- DeeToo [3]
  - Broadcasting can efficiently find things, but efficient distribution is still challenging
  - Use two rings, one transposes the other, one used for storing the other for searching
  - Broadcasts store, Distribute/aggregate searches

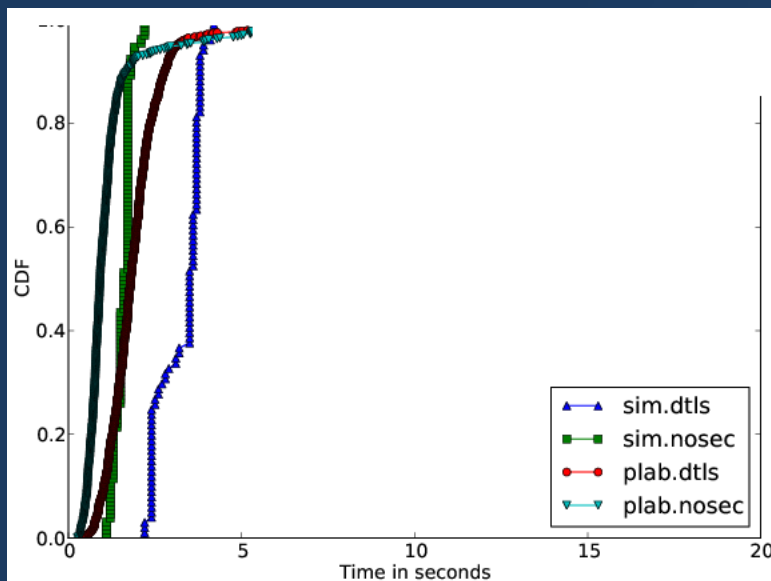
# Our Related Work

Our foray into OSNs thus far has focused on Social IP connectivity and bootstrapping private overlays

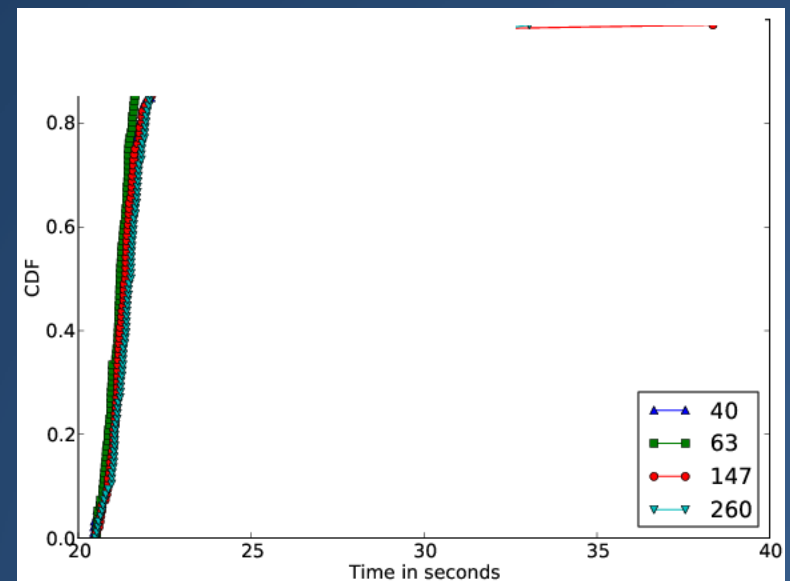
- - Lev to create IP links
  - Reuse existing network applications
  - Each peer has their own VPN address space, no IP collisions

# Our Related Work

- Creating private overlays
  - DTLS secures all P2P links
  - Certificates a CA



Add a node to an existing overlay



Bootstrap from a public overlay  
into a private overlay

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# Identity in the Directory Overlay

- Peers are identified by their PGP certificate
- Key feature
  - friends
    - Embed text data such as full name, e-mail address, and other data
  - PGP has a weak revocation model
    - Each friend signature should have a TTL
  - Friends renew signatures, keep friendship active

# Finding Friends

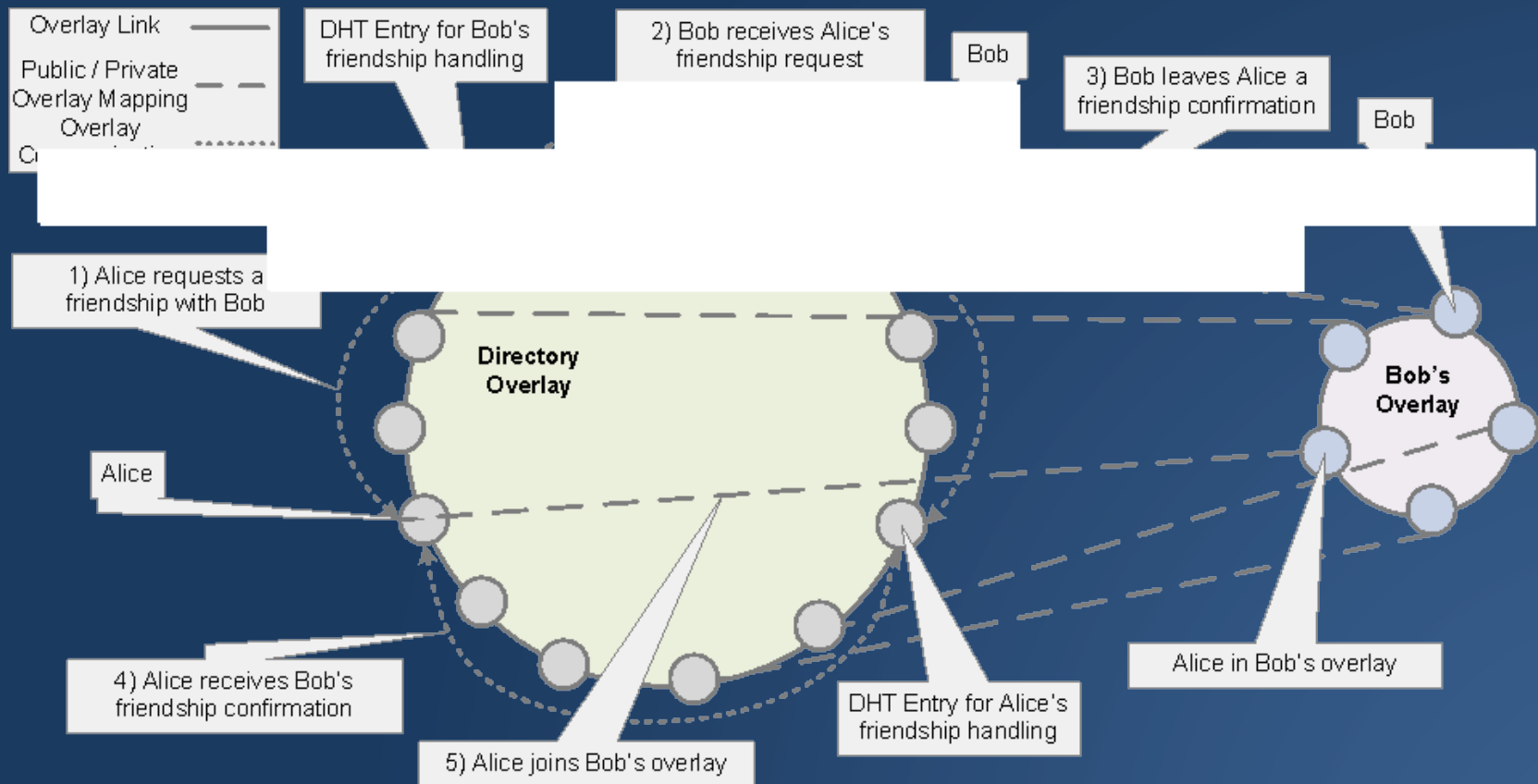
- Peers can use DeeToo to distribute and find their PGP certificates
- PGP certificates contain:

as

- Third party services can store the certificate and friendships can be established out of band
- Peers seeking extreme privacy, need not store anything inside the Directory

# Establishing a Friendship

Alice has found Bob's certificate and now is establishing a friendship



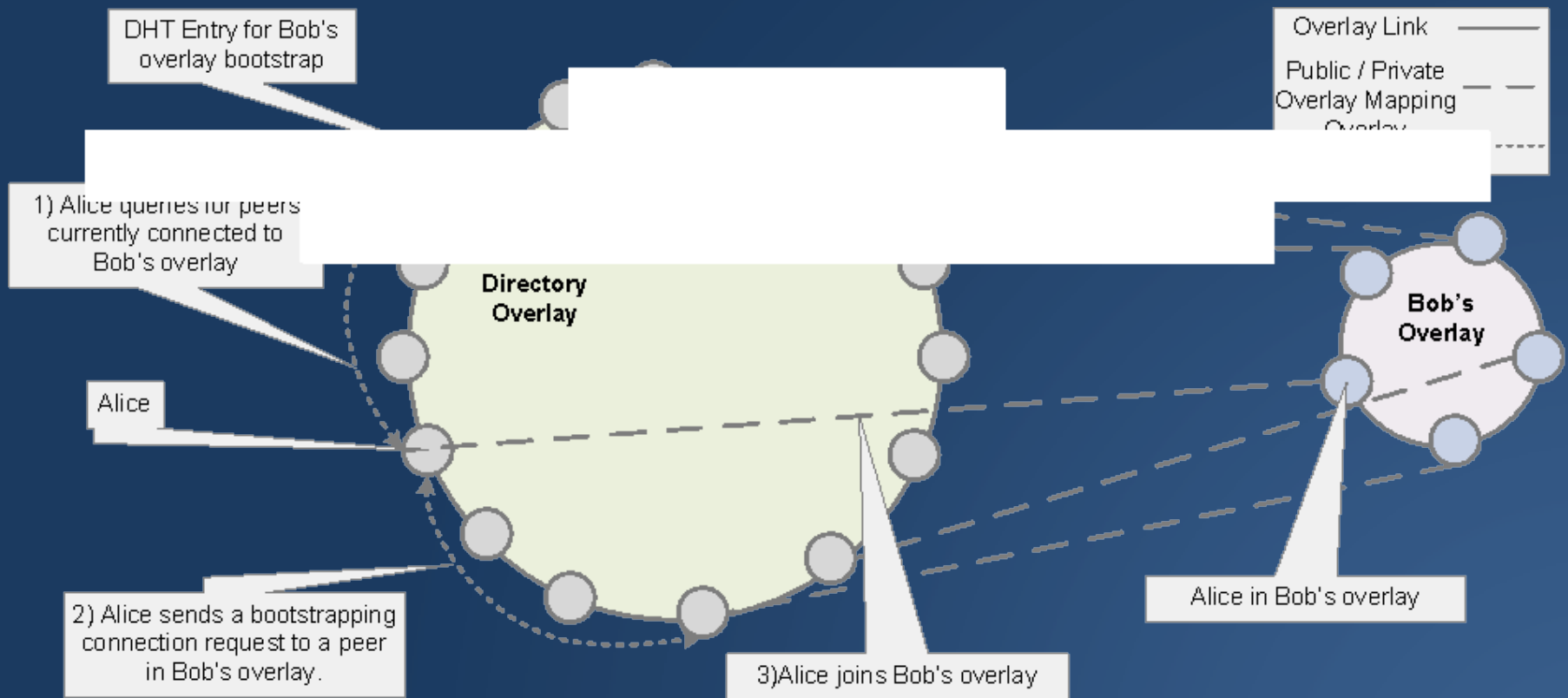


# Defriending

- Occurs if a peer does not renew a PGP signature prior to the timeout expiring
- 
- Stor HT

# Connecting to a Profile Overlay

Alice and Bob have a relationship, Alice wants to connect to Bob's overlay



# Private Messages

- Two types:
  - Friendship requests in the directory
  - Private messages in the profile overlay
- - Encrypted with sender's public key, recipient's private key
- Only peer with the private key can read message
- Message contains information such as the sender, receiver, and time of transmission

# Operations inside the Profile Overlay

- Private messages
  - Store private messages in an explicit key space
- $i$ 
  - Private message to Bob from Alice is stored in Alice's overlay
- Public messages are unencrypted but are signed to prevent unidentifiable messages

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# Remaining Challenges

- Handling Small Overlay Networks
  - Most P2P research has been performed on scalability and reliability of large (> 100,000 nodes)
  - How to efficiently leverage small structured overlays
- Overlay support for low throughput, unconnected devices
  - Passive nodes – common to many DHTs
  - Ability to modify and update users

# Remaining Challenges

- Preventing flood attacks in public overlays
  - Published d ecurity
- ap
- Data storage
  - No need to reimplement data stores for messages
  - Examples of previous work: Past / Kosh

# Related Work

The techniques in OverSoc could be used to enhance existing projects:

- PeerSon plans to integrate social networks into a
- Vis-à-vis user's node
- SafeBook relies on a central database to coordinate friendships

Challenges to getting users

bootstrapped.... Maybe Diaspora will



# Conclusion

- OverSoc techniques leverage existing structured overlay techniques to construct profile based overlays
- - Inv OSNs and integrate OverSoc concepts into them
  - Address “Remaining Challenges”

# New References

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2. J. Li, K. Sollins, and D. Lim. "Implementing Aggregation over
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# Thank you!